Dashboard & Analysis of Spotify Songs

Interactive Visualisation and Data Analysis Project

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Goal

- Examined a real-world Spotify collection (more than 100000 tracks).
- Examined trends in the genres, audio features, and popularity of tracks.
- Used Python to visualise insights with pandas, matplotlib, seaborn, and plotly.
- Used Streamlit to create an interactive dashboard.

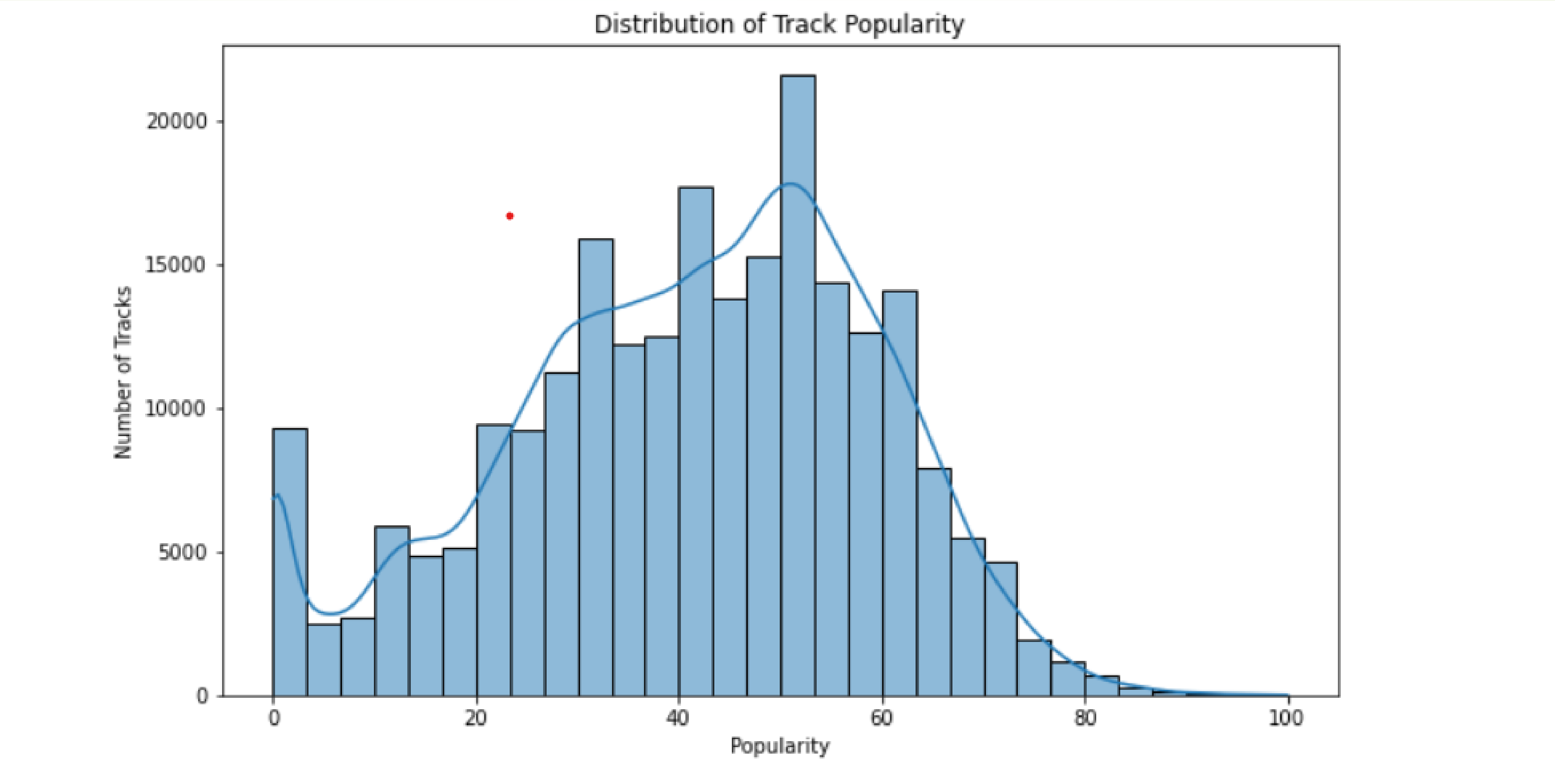
Overview of the Dataset

- SpotifyFeatures.csv is the dataset.
- More than 100000 tracks
- Important columns include danceability, energy, valence, tempo, popularity, artist and track names, and genre.
- Kaggle is the source.

	genre	artist_name	track_name	track_id	popularity	acousticness	danceability	duration_ms	energy	instrumentalness	key	liveness	loudness	mode	speechiness	tempo	time_signature	valence
0	Movie	Henri Salvador	C'est beau de faire un Show	0BRjO6ga9RKCKjfDqeFgWV	0	0.611	0.389	99373	0.910	0.000	C#	0.3460	-1.828	Major	0.0525	166.969	4/4	0.814
1	Movie	Martin & les fées	Perdu d'avance (par Gad Elmaleh)	0BjC1NfoEOOusryehmNudP	1	0.246	0.590	137373	0.737	0.000	F#	0.1510	-5.559	Minor	0.0868	174.003	4/4	0.816
2	Movie	Joseph Williams	Don't Let Me Be Lonely Tonight	0CoSDzoNIKCRs124s9uTVy	3	0.952	0.663	170267	0.131	0.000	C	0.1030	-13.879	Minor	0.0362	99.488	5/4	0.368
3	Movie	Henri Salvador	Dis-moi Monsieur Gordon Cooper	0Gc6TVm52BwZD07Ki6tIvf	0	0.703	0.240	152427	0.326	0.000	C#	0.0985	-12.178	Major	0.0395	171.758	4/4	0.227
4	Movie	Fabien Nataf	Ouverture	0IuslXpMROHdEPvSI1fTQK	4	0.950	0.331	82625	0.225	0.123	F	0.2020	-21.150	Major	0.0456	140.576	4/4	0.390

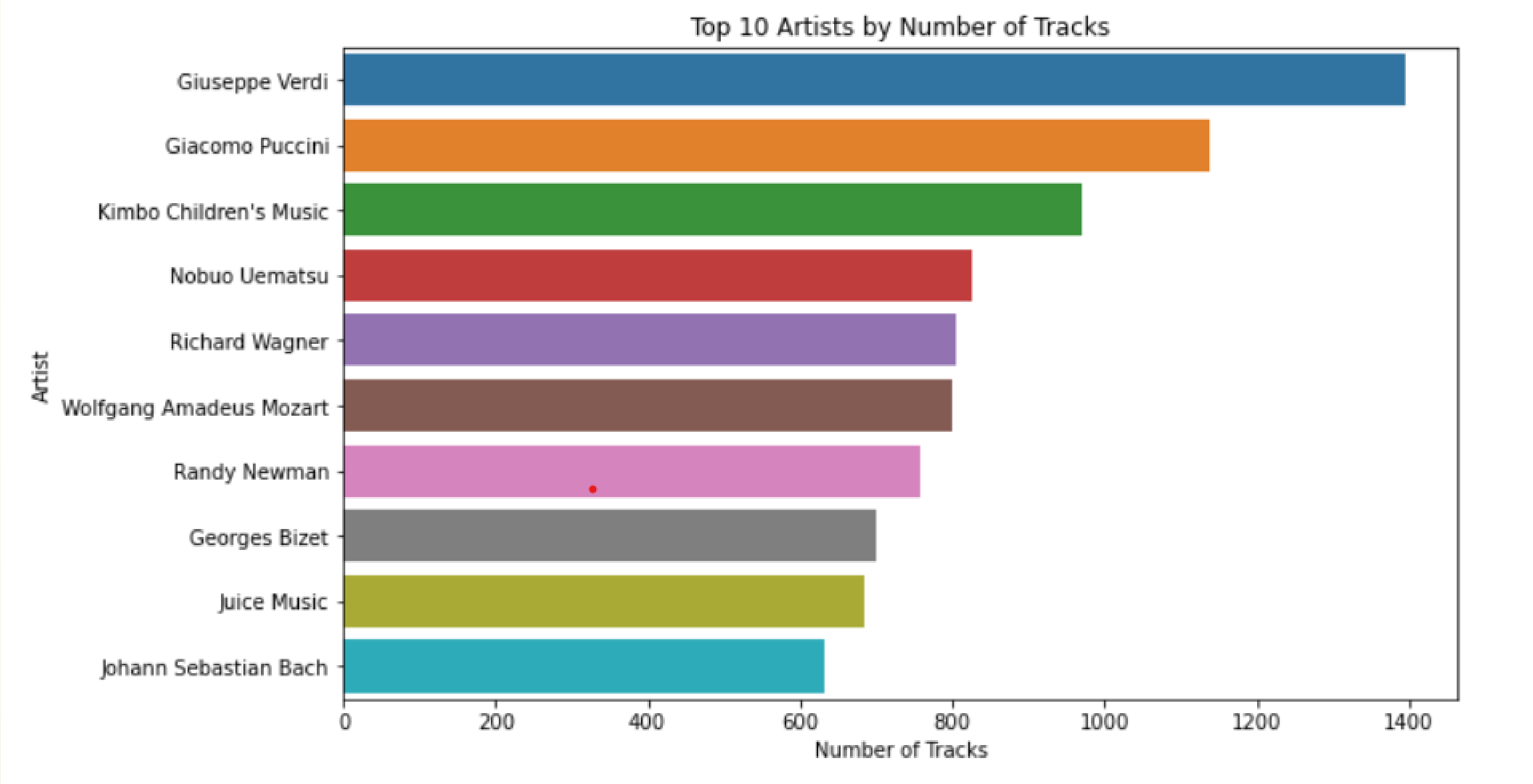
Distribution of Popularity

- Observation: The majority of songs are between 40 and 70 percent popular; very few are very popular or unpopular.



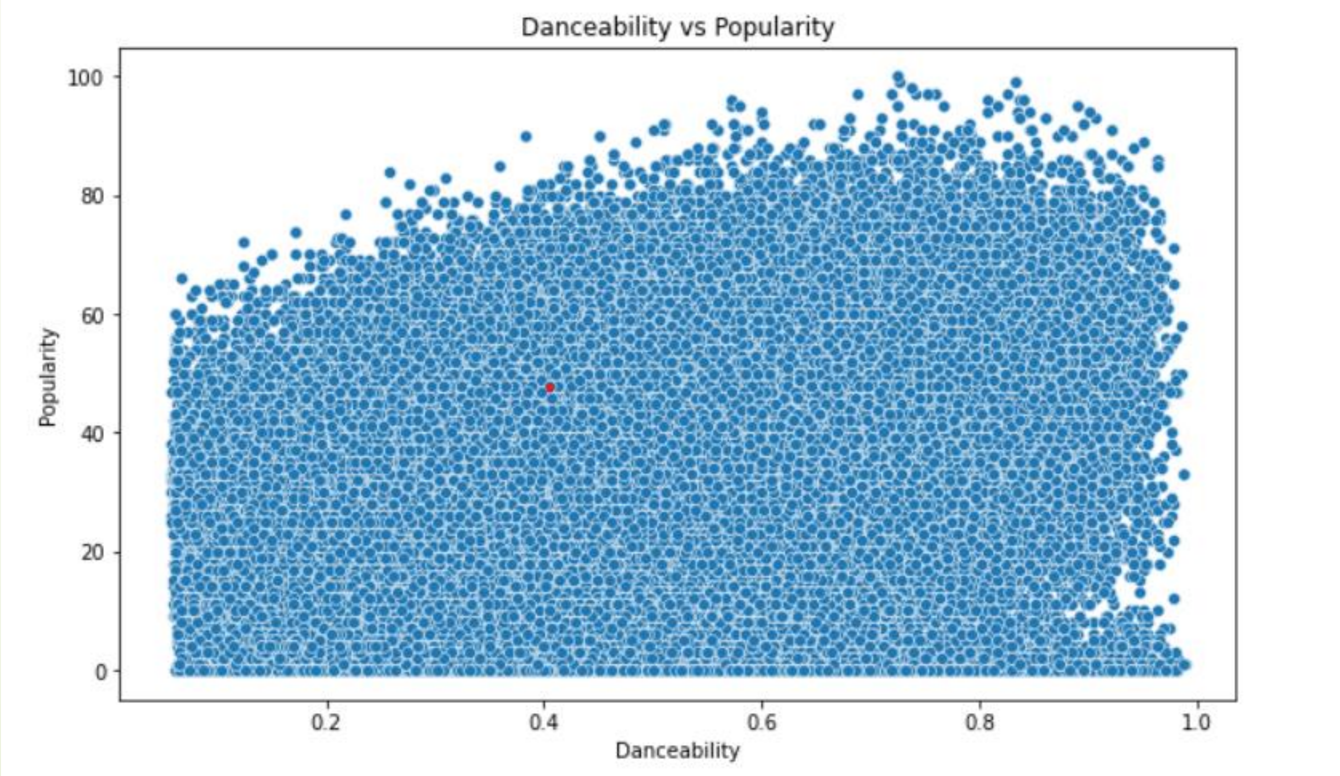
Artists with the Most Tracks

- Question for Analysis: Which musicians have the most tracks?
- Conclusion: The most productive artists in the dataset are the ones with the most tracks.



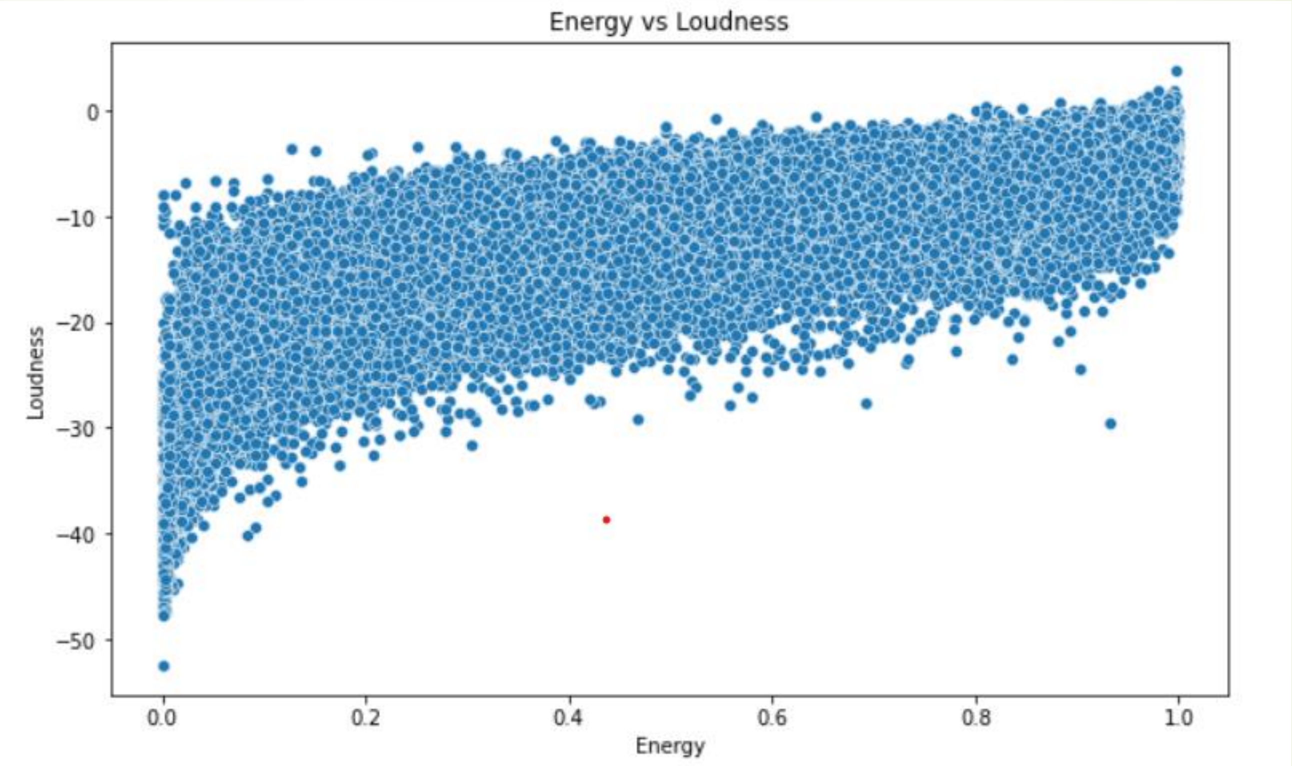
Popularity vs. Danceability

- Analytical Question: What is the relationship between popularity and danceability?
- Observation: There is a weak correlation between tracks with higher danceability and slightly higher popularity.



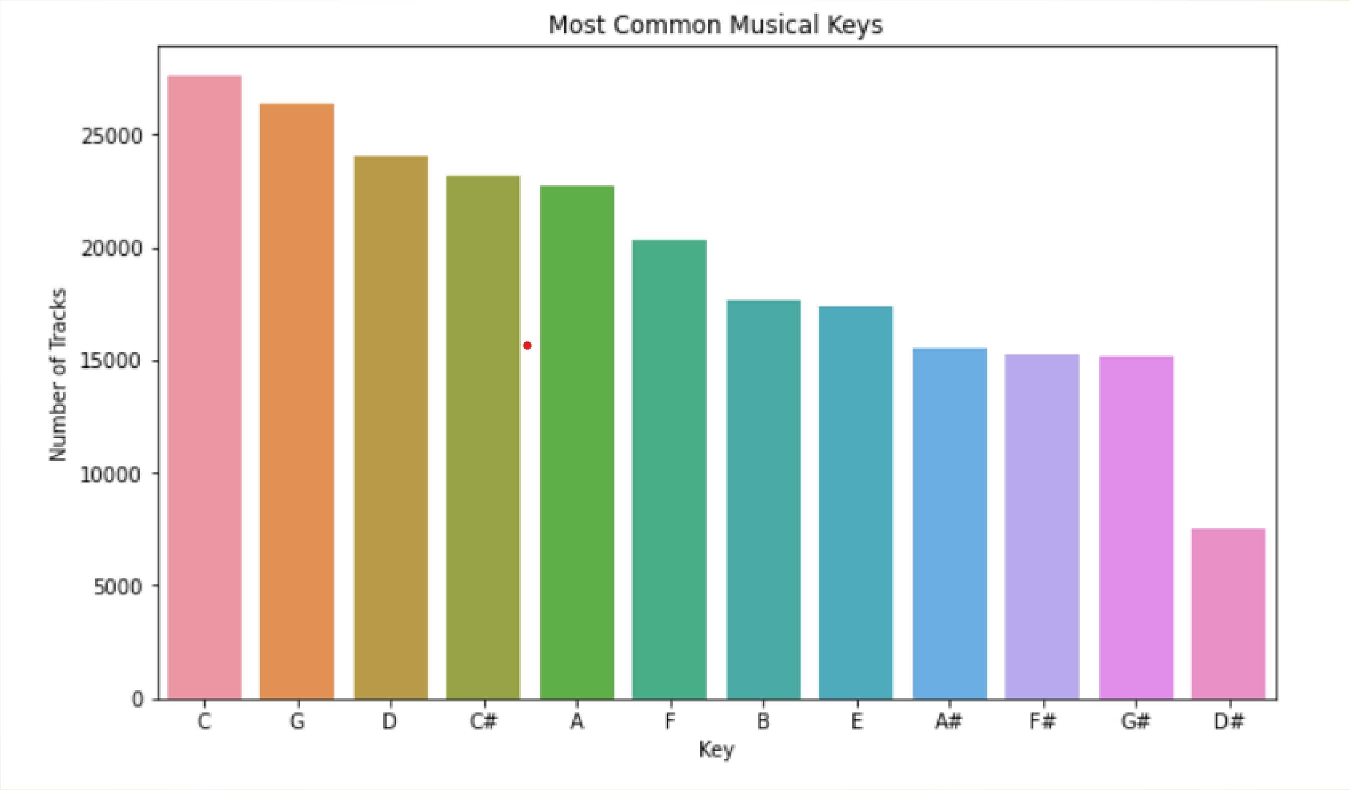
Popularity vs. Energy

- Analytical Question: Does popularity change with increased energy?
- The popularity of energetic tracks is higher, indicating that listeners prefer them.



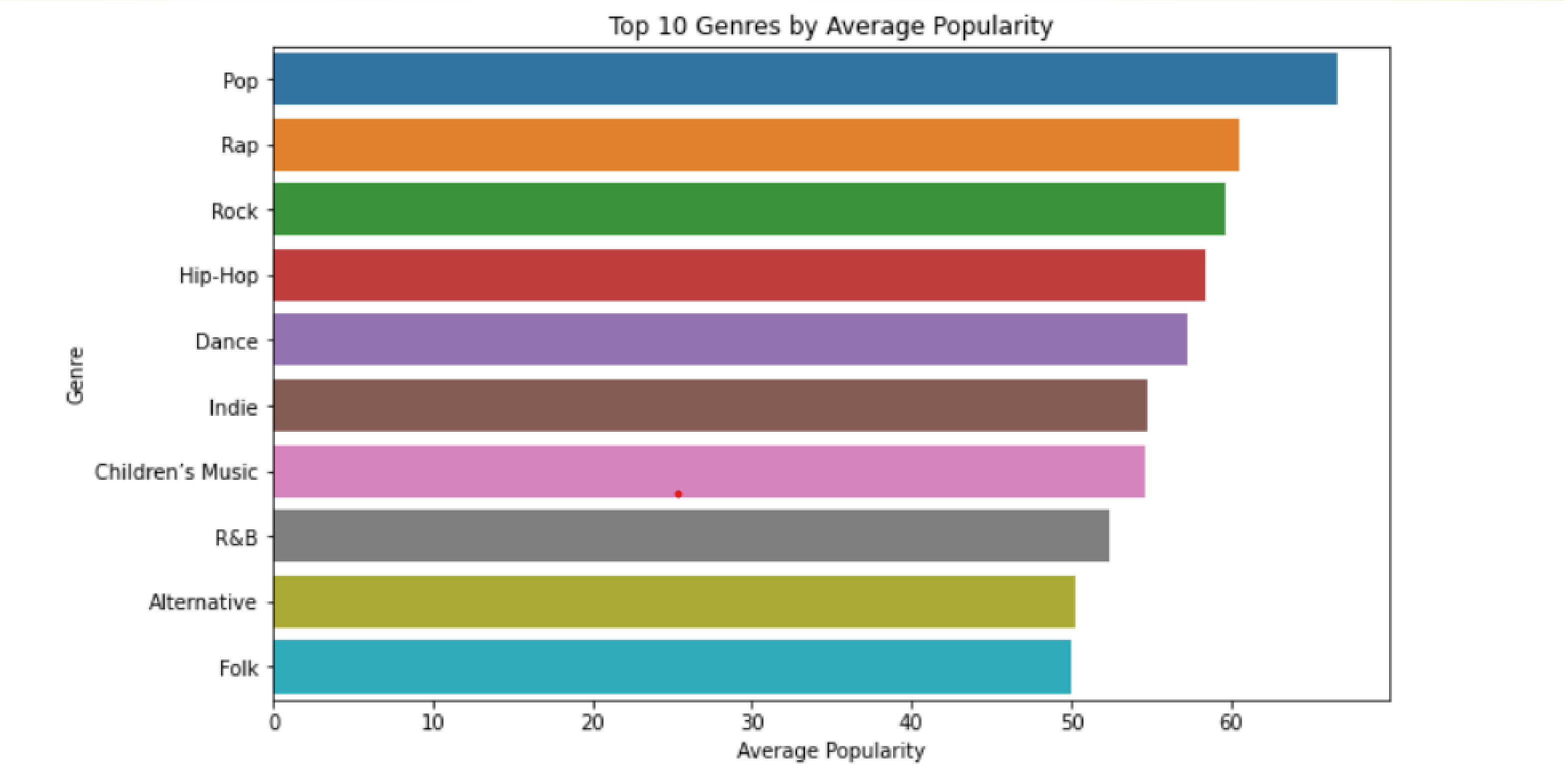
The Most Frequently Used Musical Keys

- Analytical Question: What are the most commonly used keys?
- Observation: The most frequently used keys are C (0) and G (7), which correspond to common composition trends.



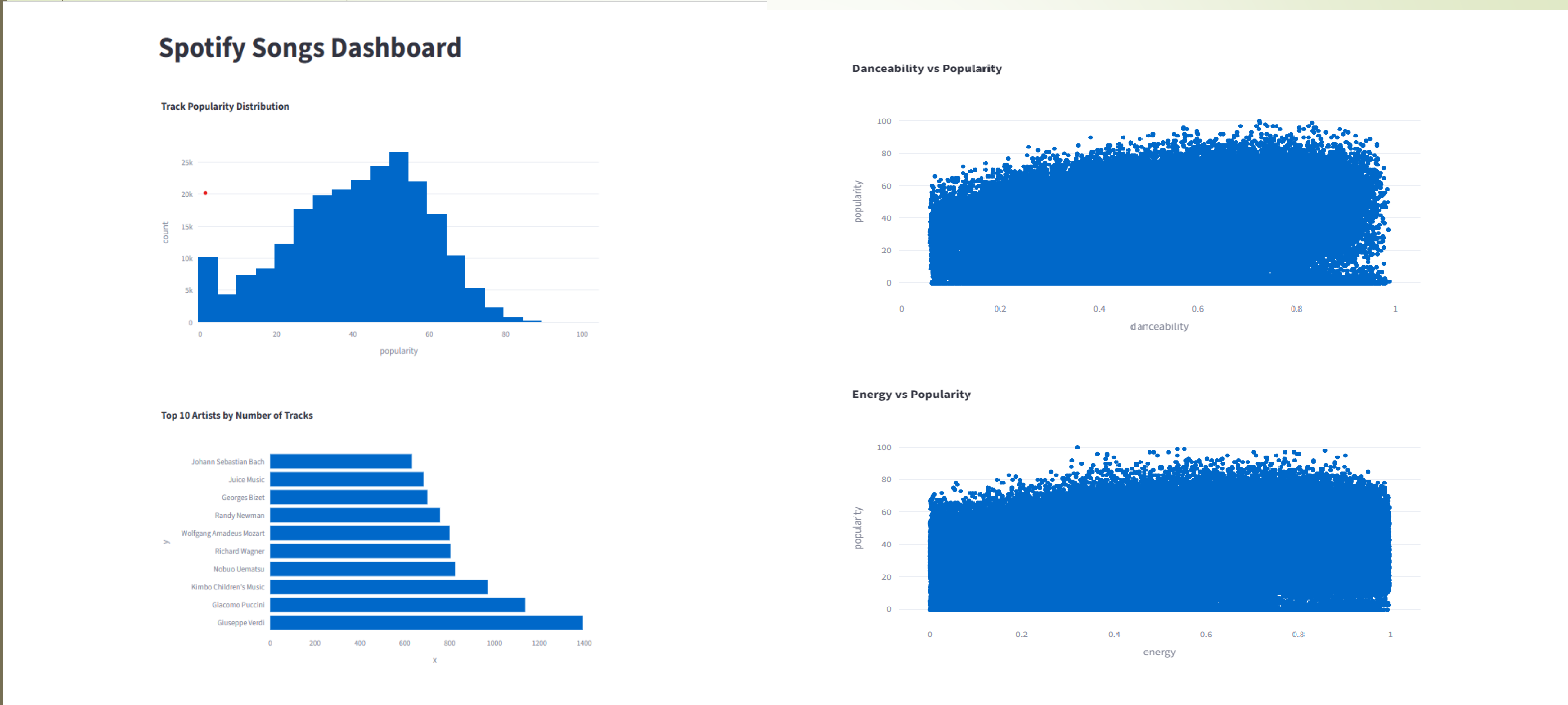
Average Popularity of the Top 10 Genres

- Analytical Question: What are the most popular genres?
- Insight: Listener preferences are reflected in the dominance of pop and hip-hop in average popularity.



Dashboard Streamlight

- Streamlit + Plotly was used to create the interactive dashboard enables users to dynamically explore top artists, audio features, and track popularity.



Important Lessons Learned

- The majority of songs are moderately popular; very few are huge hits or failures.
- Songs that are lively and somewhat danceable are typically more well-liked.
- The Spotify dataset is dominated by particular artists and genres.
- Trends can be explored interactively with the help of the streamlined dashboard.

Utilised Tools and Libraries

- Python libraries: seaborn, plotly, pandas, matplotlib, and streamlit
- IDEs: Jupyter Notebook and Visual Studio Code
- Dataset: Spotify Features on Kaggle
- Github Link: https://github.com/NakshatraGokule/Spotify_Project
- Streamlit: https://nakshatragokule-spotify-project-app-pgljkh.streamlit.app/

Sources of References and Datasets

- Spotify Dataset: Features of Kaggle Spotify
- Documentation for Python Libraries
- Streamlit Community Cloud